

Gerçek Veri Baseline Eğitim Raporu

is_missing flags + Median Imputation -- 3 Pipeline, 7 Model

Tarih: 2026-05-12 11:43

Eğitim verisi: YARISMA_TRAIN_MASTER.csv

Test panelleri: KANSER, PAH, CFTR

Pipeline A (OHE) : LightGBM, XGBoost, SVM

Pipeline B (LE) : LightGBM, XGBoost, SVM

Pipeline C (CatBoost): CatBoost (Ordered Target Statistics)

Ham veri: 2931 satır x 352 sütun (Label dahil)

Yüksek NaN (>%50) sütunlar: 163 -> is_missing flag olarak tutuldu

Düşük NaN (<=50%) sütunlar: 180 -> MASTER train mediani ile dolduruldu

1. On Isleme Stratejisi

NaN Yonetimi

Klasik %50 NaN drop stratejisi terk edildi. CLAUDE.md'a gore eksiklik bilgi tasiyor: MASTER'da Label=1 satirlarda eksiklik %59.9, Label=0'da %41.2. Bu fark, eksikligi bir sinyal olarak modele sunmanin gecerililigini gosteriyor. NaN>%50 olan sutunlar drop edilmek yerine is_missing_* binary flag olarak tutuluyor; sutunun orijinal degeri de (varsa) ayni sekilde matriste kalıyor. NaN>%50 olan sutunun kendisi icin de median imputation uygulanir (model NaN kabul etmez).

Median vs Mean Imputation

AL_ sutunlari populasyon freakans ve patojenite skoru icerir. Bu dagilimlar genellikle saga carpik (cogu varyant nadir). Mean imputation, carpik dagilimda aykiri gozlemlere karsi hassastir. Median bu etkiye karsi direncli oldugu icin tercih edildi.

Leakage Korumalari

Tum fit islemleri (median hesabi, OHE, LabelEncoder) YALNIZCA MASTER train bolumu uzerinde yapildi. Test seti ve panellere yalnizca transform uygulanir. Panel testlerinde MASTER train'dan cikarilan median ve encoderlar kullanilir.

2. Veri Ozeti

MASTER ham boyut: 2931 satir x 352 sutun

Label: Pozitif=2149, Negatif=782 (oran=0.733)

Kategorik sutunlar (8): CAT_1, CAT_2, CAT_3, CAT_4, CAT_5, CAT_6, AA_1, AA_2

Sayisal sutun toplam: 343

NaN > %50 -> is_missing flag: 163

NaN <= %50 -> median imputed: 180

OHE boyutu: 618 ozellik

RAW boyutu: 514 ozellik

Hold-out: egitim=2344, test=587

Panel Verileri:

Panel	Toplam	Pozitif	Negatif	MASTER Ortusme
KANSER	388	268	120	246/388 (63%)
PAH	372	310	62	255/372 (69%)
CFTR	111	90	21	77/111 (69%)

3. MASTER Hold-Out Sonuclari (%20 hold-out)

Pipeline A -- OHE

Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Bal.Acc	Sec
lgbm_ohe	0.8932	0.8334	0.9136	0.8402	0.9535	0.5398	0.7283	24
xgb_ohe	0.8877	0.8388	0.9257	0.8431	0.9372	0.5231	0.7298	11
svm_ohe	0.8749	0.7949	0.9119	0.8099	0.9512	0.4337	0.6698	33

Pipeline B -- LE

Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Bal.Acc	Sec
lgbm_le	0.8940	0.8415	0.9207	0.8508	0.9419	0.5535	0.7448	22
xgb_le	0.8832	0.8298	0.9197	0.8191	0.9581	0.4785	0.6893	11
svm_le	0.8631	0.7810	0.8993	0.7790	0.9674	0.3334	0.6079	20

Pipeline C -- CatBoost

Model	F1	AUC-ROC	AUC-PR	Prec	Recall	MCC	Bal.Acc	Sec
catboost	0.8940	0.8364	0.9083	0.8433	0.9512	0.5456	0.7335	52

En iyi Model (F1 bazında):

[LE] lgbm_le -- F1=0.8940 AUC=0.8415 Prec=0.8508 Recall=0.9419

4. Panel Test Sonuclari

Her panel dosyasinin tum satirlari test edildi. MASTER train median ve encoder kullanildi (fit yok).

Panel	Sec	Model	F1	AUC-ROC	Prec	Recall	MCC	N
CFTR	CATBOOST	catboost	0.9392	0.9429	0.9341	0.9444	0.6713	111
CFTR	LE	lgbm_le	0.9282	0.9328	0.9231	0.9333	0.6114	111
CFTR	LE	svm_le	0.9175	0.8270	0.8558	0.9889	0.4425	111
CFTR	LE	xgb_le	0.9405	0.9111	0.9158	0.9667	0.6531	111
CFTR	OHE	lgbm_ohe	0.9444	0.9593	0.9444	0.9444	0.7063	111
CFTR	OHE	svm_ohe	0.9140	0.7101	0.8854	0.9444	0.4819	111
CFTR	OHE	xgb_ohe	0.9274	0.9280	0.9326	0.9222	0.6254	111
KANSER	CATBOOST	catboost	0.9036	0.9000	0.8266	0.9963	0.6555	388
KANSER	LE	lgbm_le	0.9072	0.9254	0.8408	0.9851	0.6687	388
KANSER	LE	svm_le	0.8679	0.8917	0.7710	0.9925	0.4921	388
KANSER	LE	xgb_le	0.8829	0.9105	0.8000	0.9851	0.5640	388
KANSER	OHE	lgbm_ohe	0.9016	0.8964	0.8392	0.9739	0.6457	388
KANSER	OHE	svm_ohe	0.8833	0.9023	0.7982	0.9888	0.5662	388
KANSER	OHE	xgb_ohe	0.8962	0.8884	0.8355	0.9664	0.6244	388
PAH	CATBOOST	catboost	0.9240	0.7789	0.8736	0.9806	0.4111	372
PAH	LE	lgbm_le	0.9294	0.8318	0.8860	0.9774	0.4768	372
PAH	LE	svm_le	0.9055	0.7631	0.8459	0.9742	0.1650	372
PAH	LE	xgb_le	0.9164	0.7913	0.8528	0.9903	0.2858	372
PAH	OHE	lgbm_ohe	0.9136	0.8101	0.8757	0.9548	0.3588	372
PAH	OHE	svm_ohe	0.8962	0.6399	0.8394	0.9613	0.0748	372
PAH	OHE	xgb_ohe	0.9198	0.8046	0.8661	0.9806	0.3594	372

En iyi Model (Panel Bazinda):

KANSER: [LE] lgbm_le -- F1=0.9072 Recall=0.9851 AUC=0.9254

PAH: [LE] lgbm_le -- F1=0.9294 Recall=0.9774 AUC=0.8318

CFTR: [OHE] lgbm_ohe -- F1=0.9444 Recall=0.9444 AUC=0.9593

5. Cross-Model Error Analizi

Hold-out satirlari: 587

Tum modeller yanlis (score=7): 61 satir

Cogunluk yanlis (score>=4): 104 satir

Tum modeller dogru (score=0): 414 satir

Difficulty Score Dagilimi:

Score=0: 414 satir

Score=1: 43 satir

Score=2: 16 satir

Score=3: 10 satir

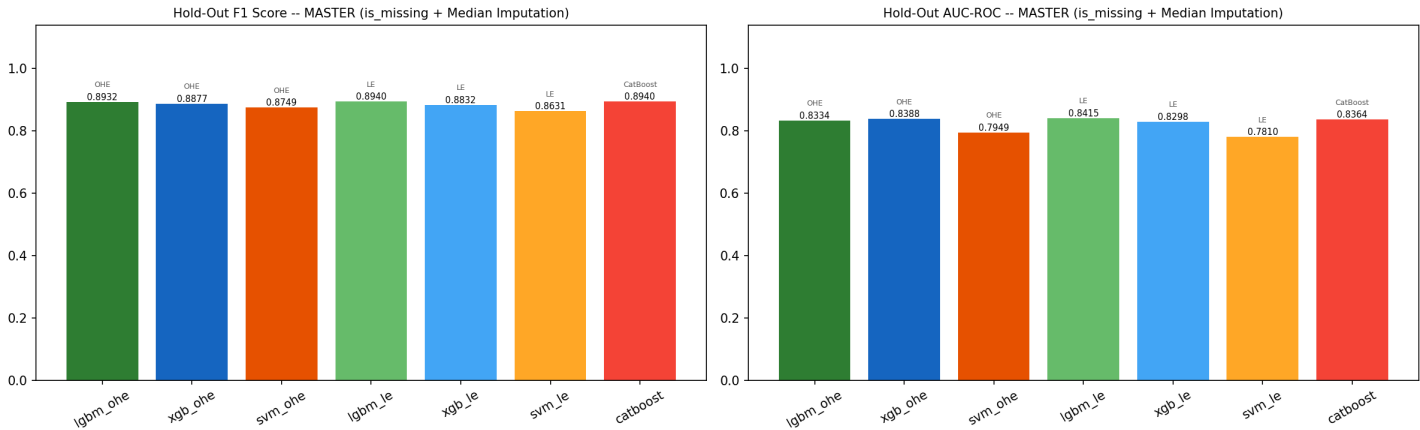
Score=4: 13 satir

Score=5: 13 satir

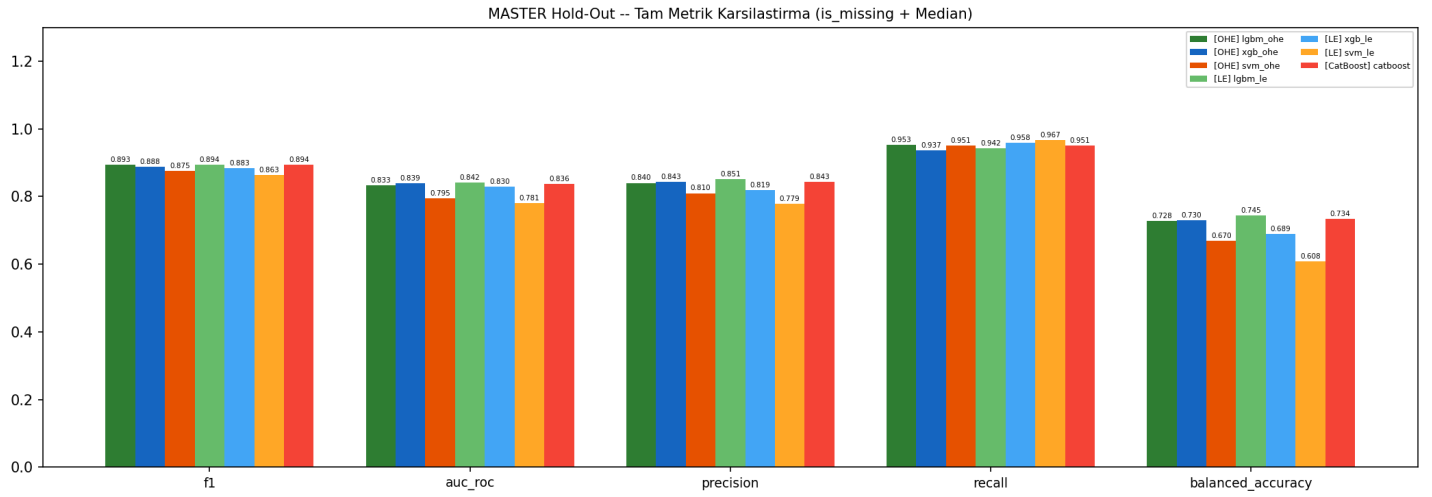
Score=6: 17 satir

Score=7: 61 satir

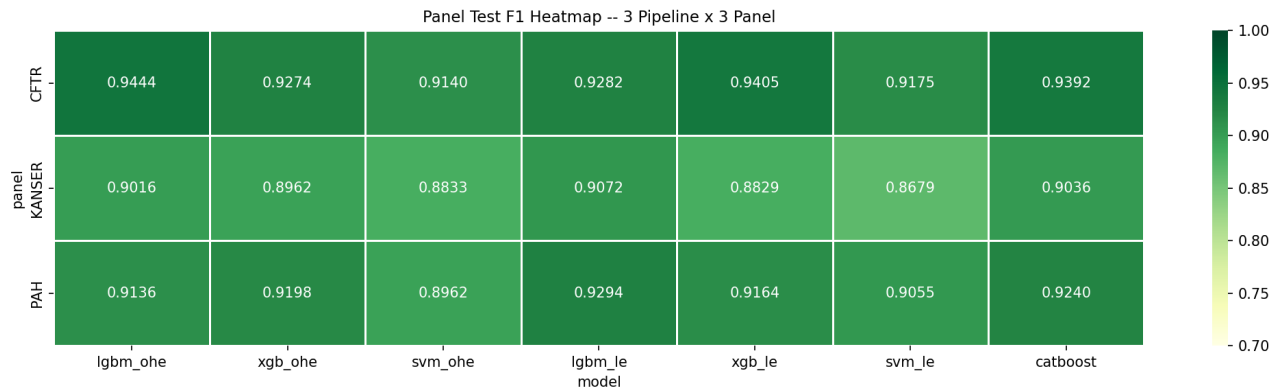
Master F1 Auc



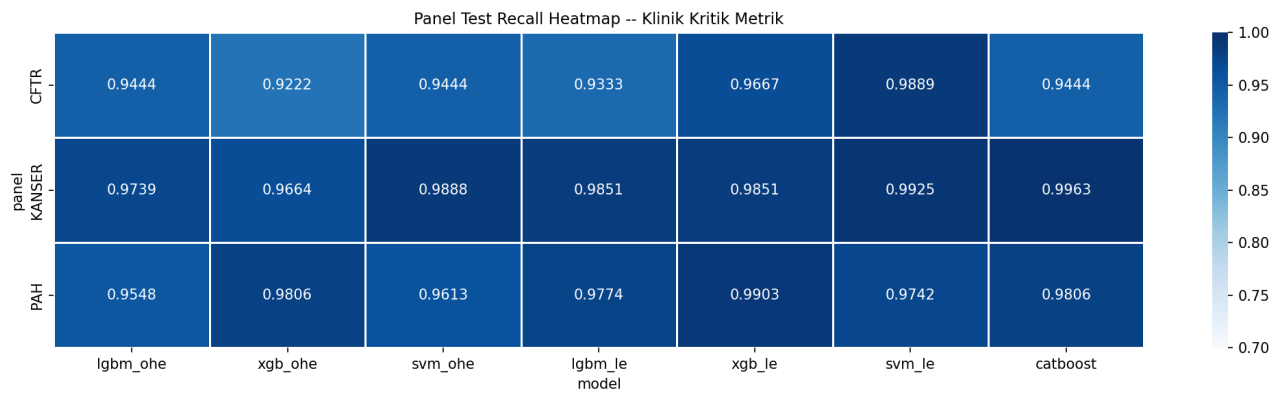
Master Metrics



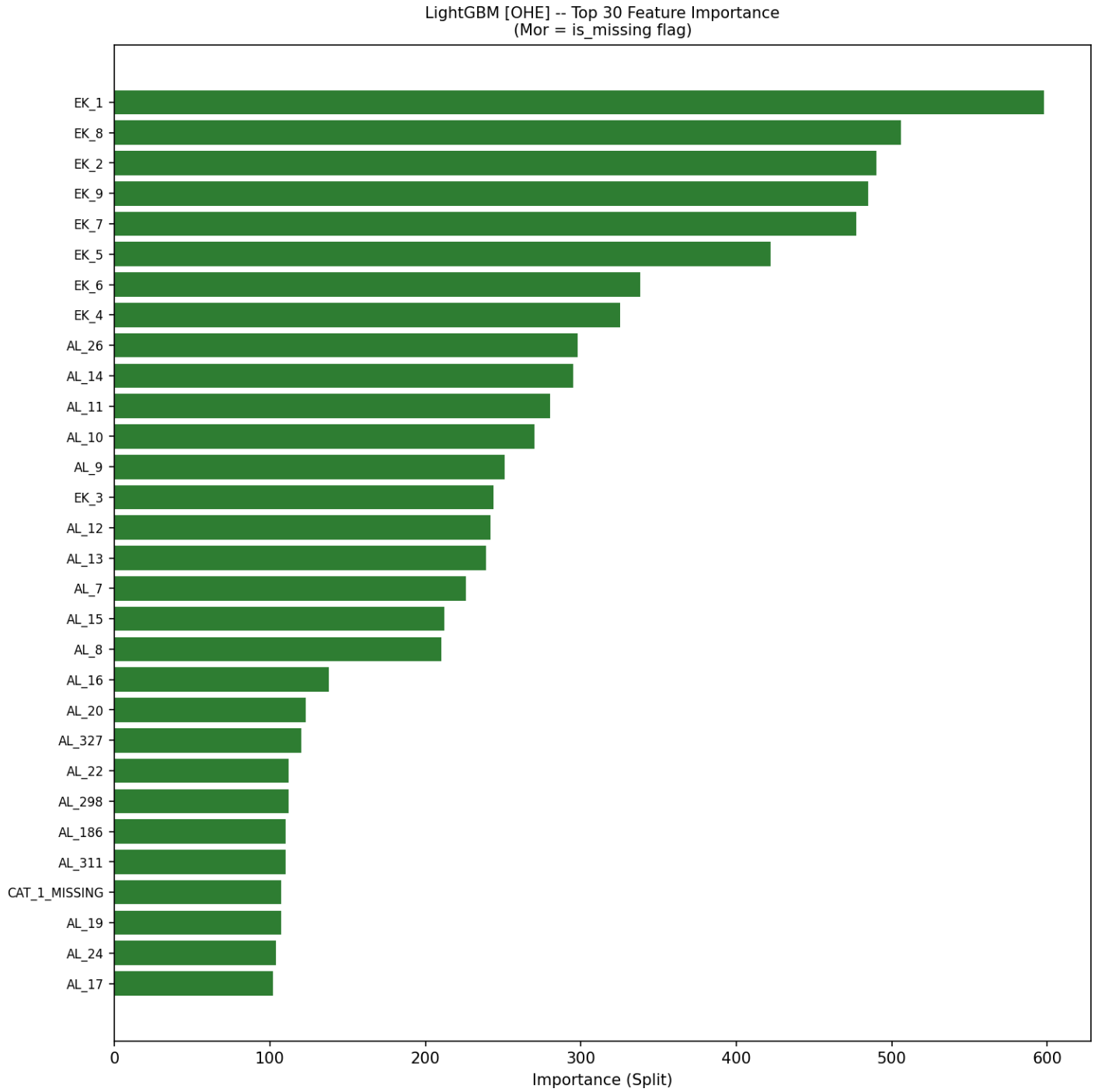
Panel F1 Heatmap

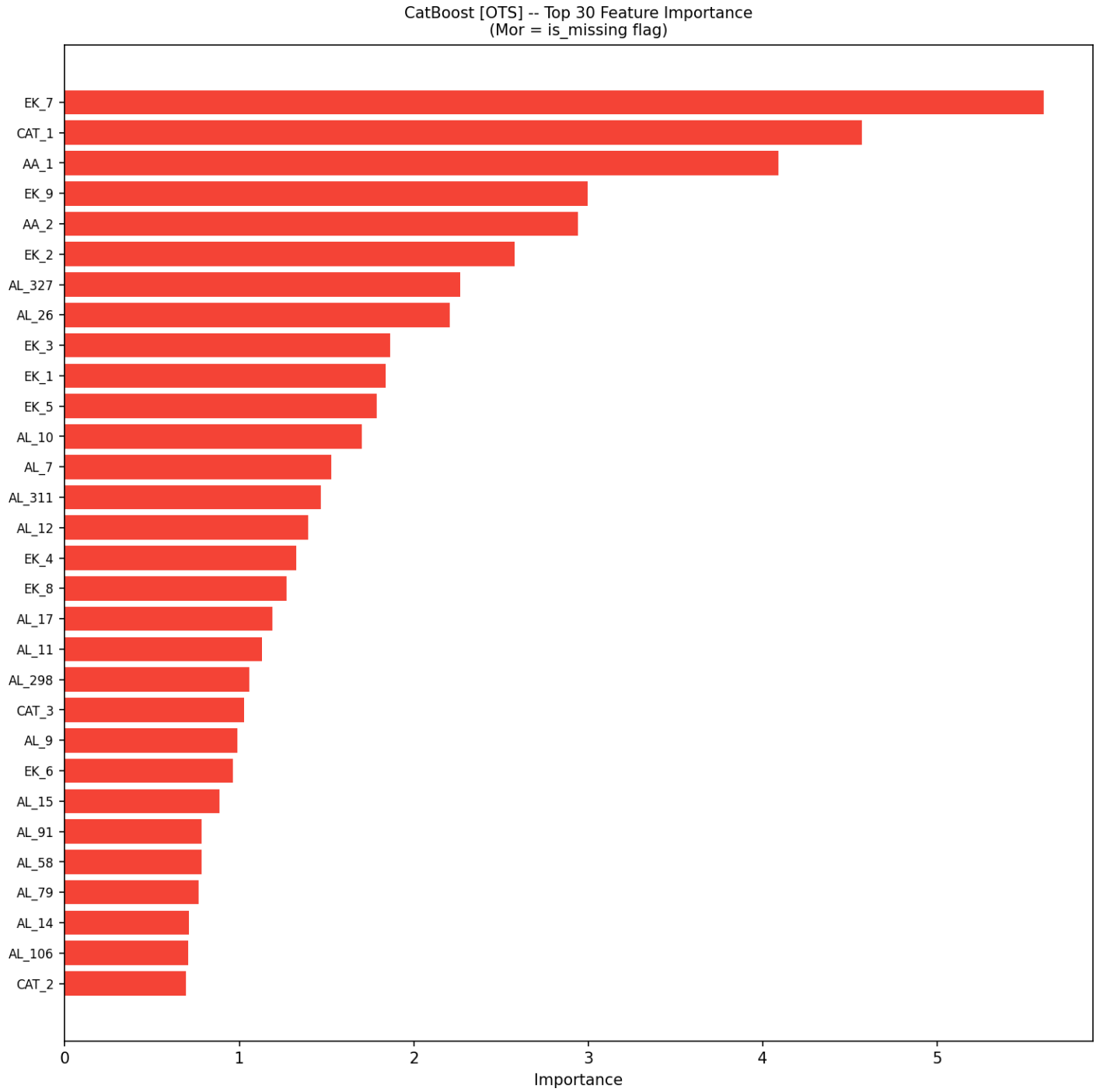


Panel Recall Heatmap

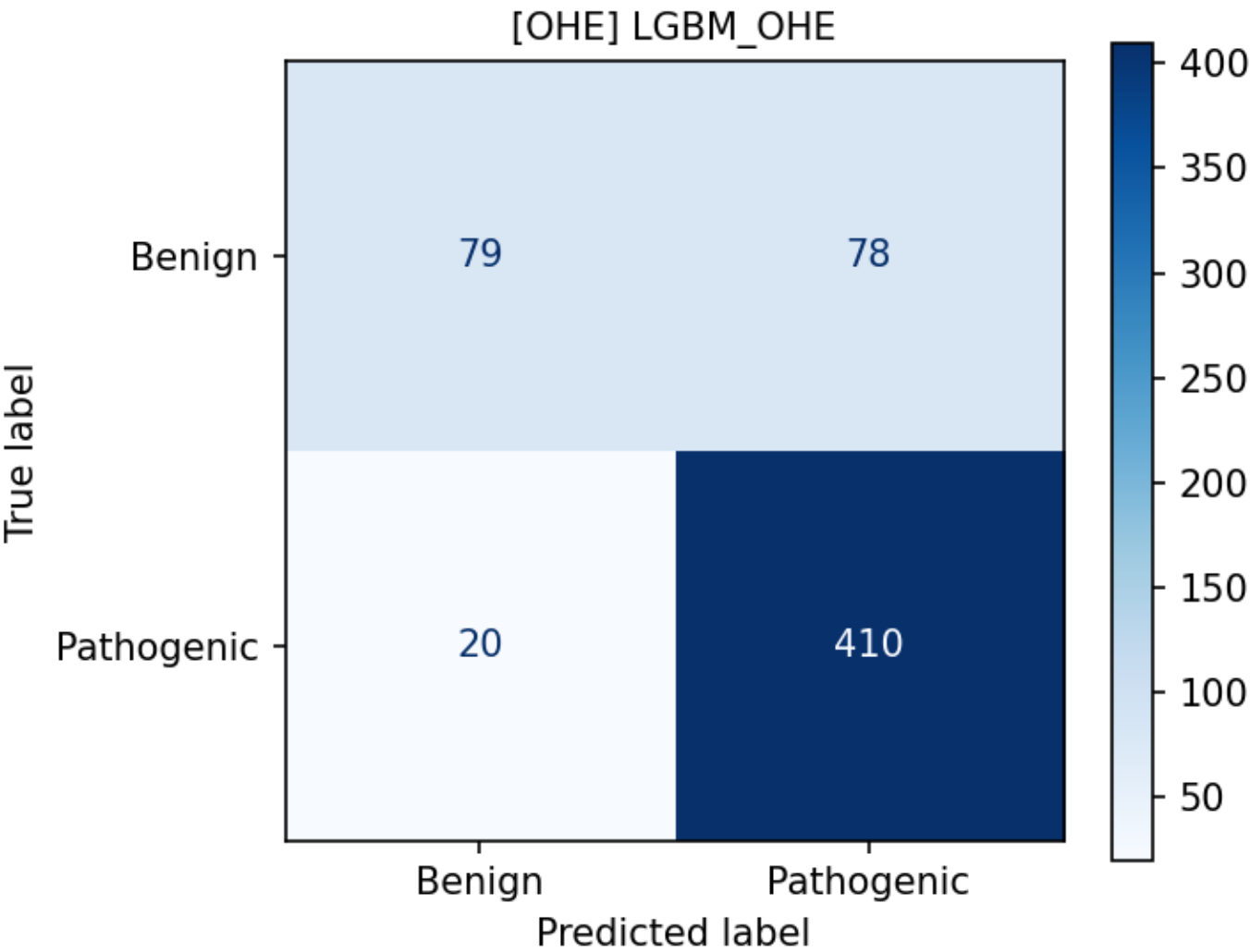


Lgbm Ohe Feature Importance

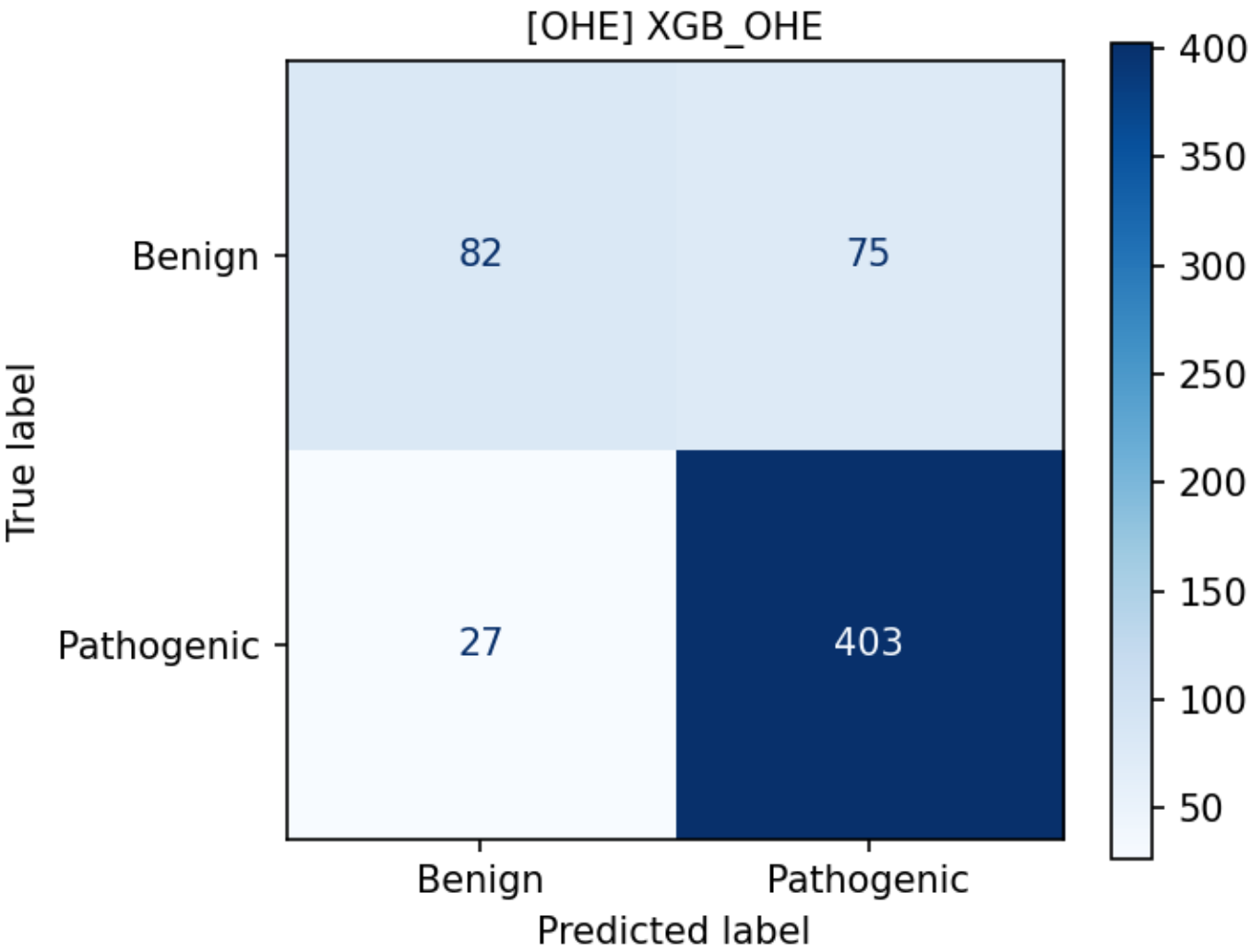


Catboost Feature Importance

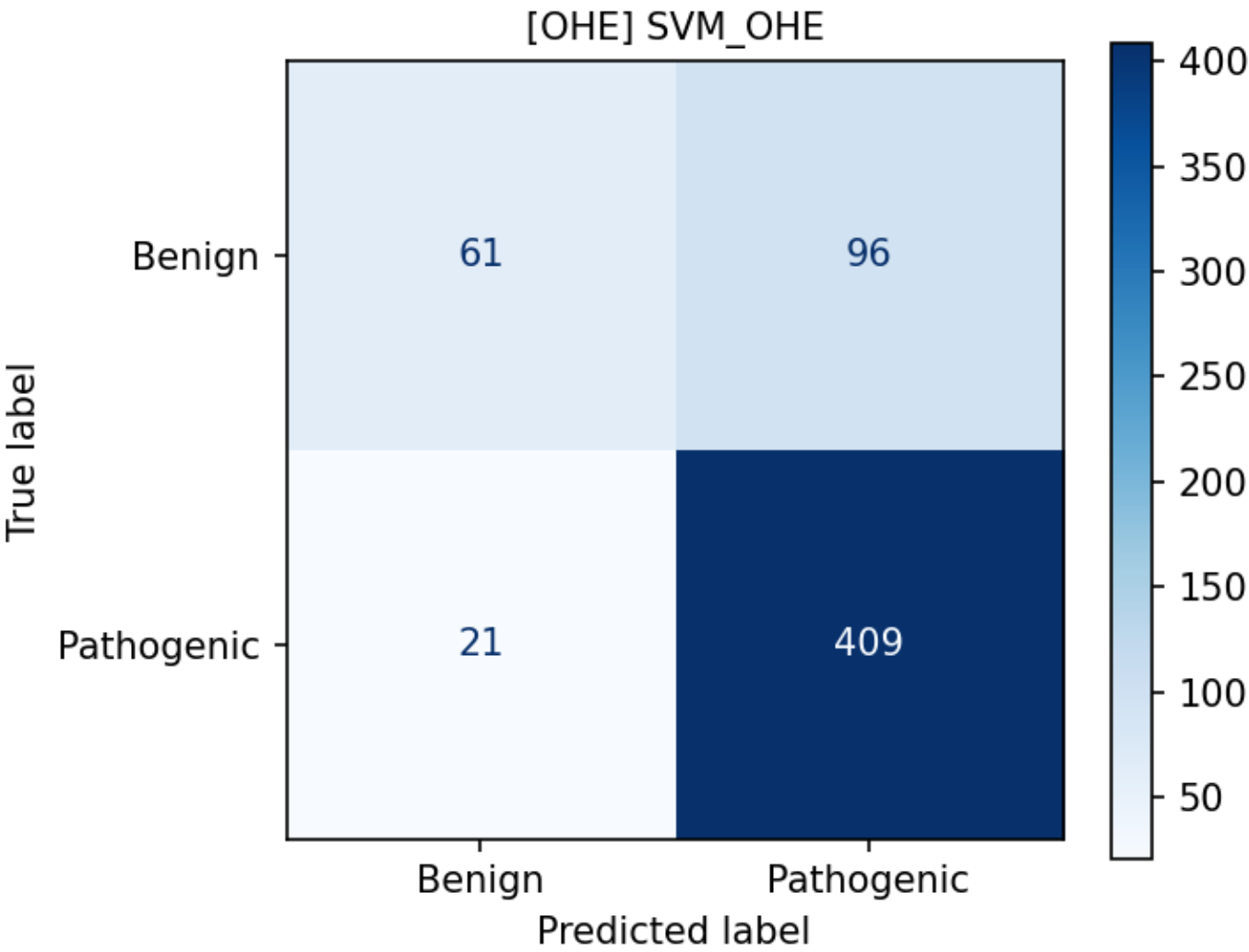
Master Lgbm Ohe Cm



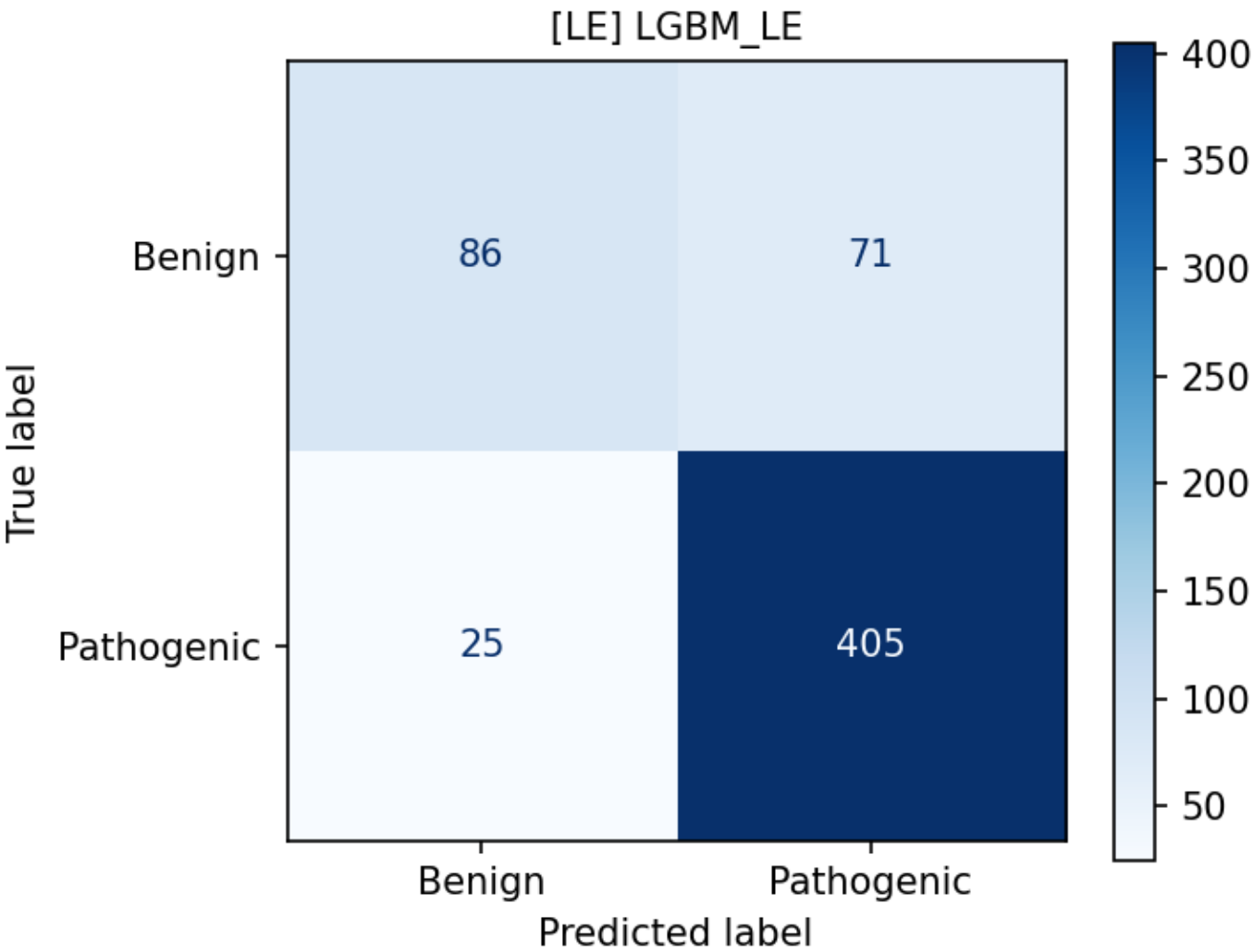
Master Xgb Ohe Cm



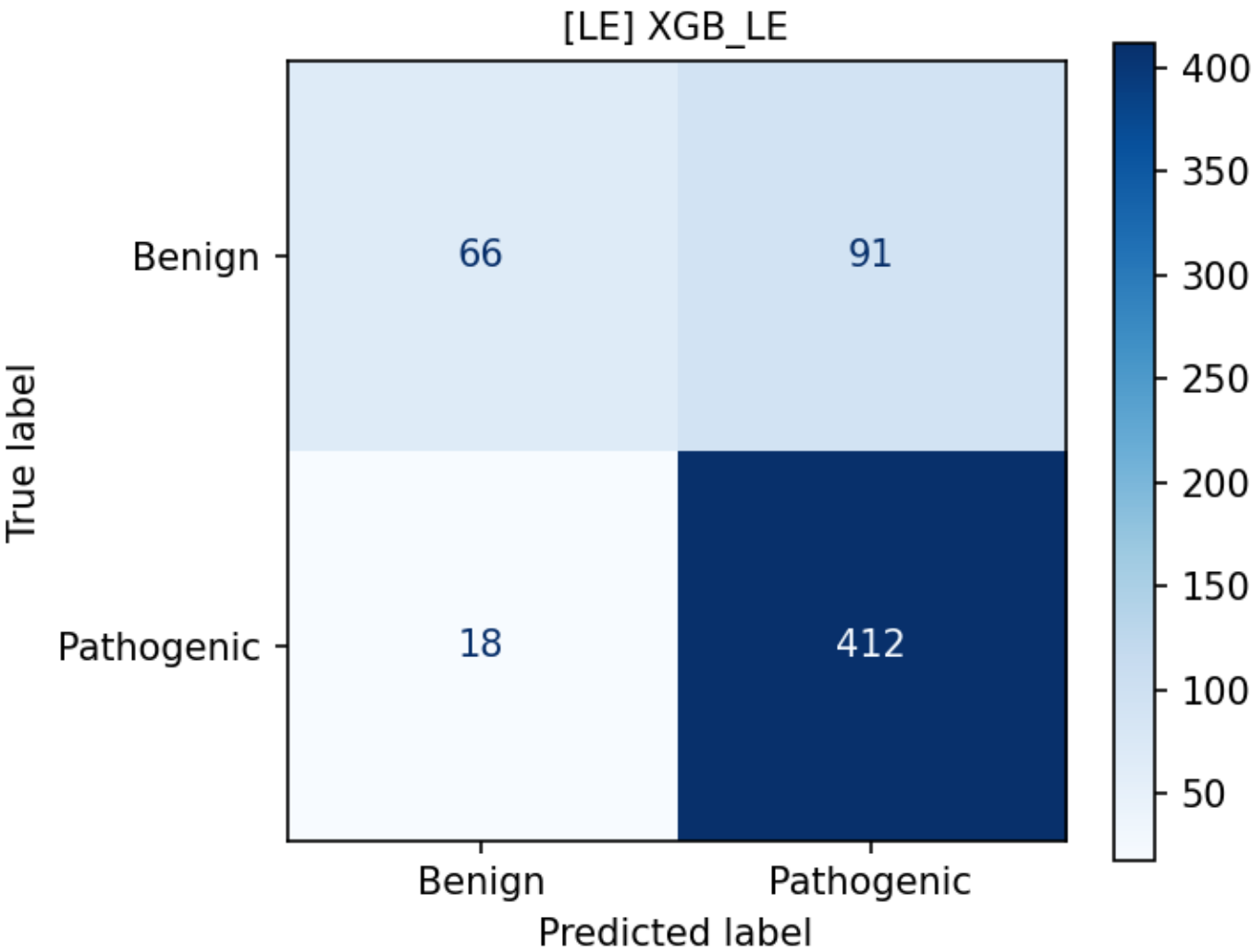
Master Svm Ohe Cm



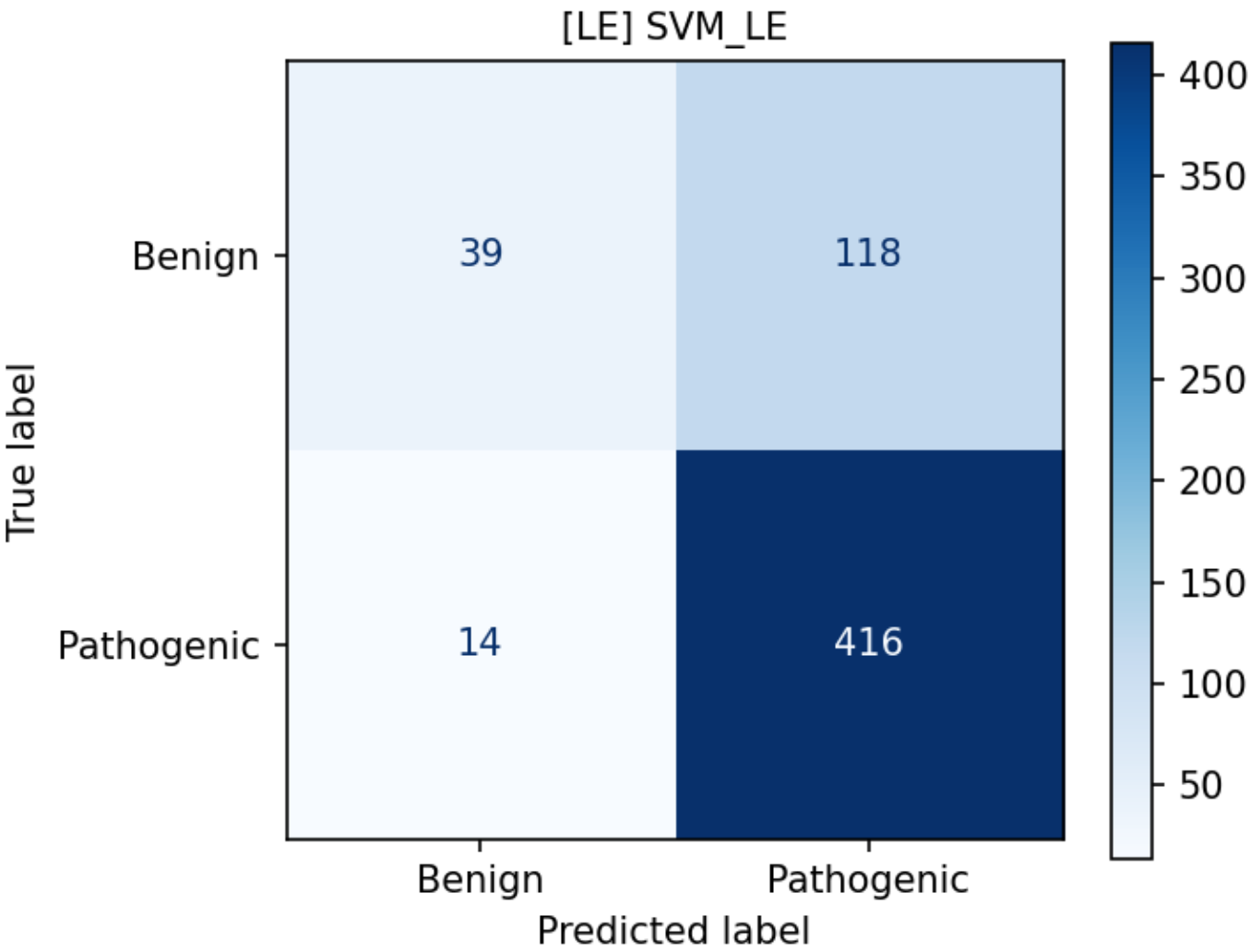
Master Lgbm Le Cm



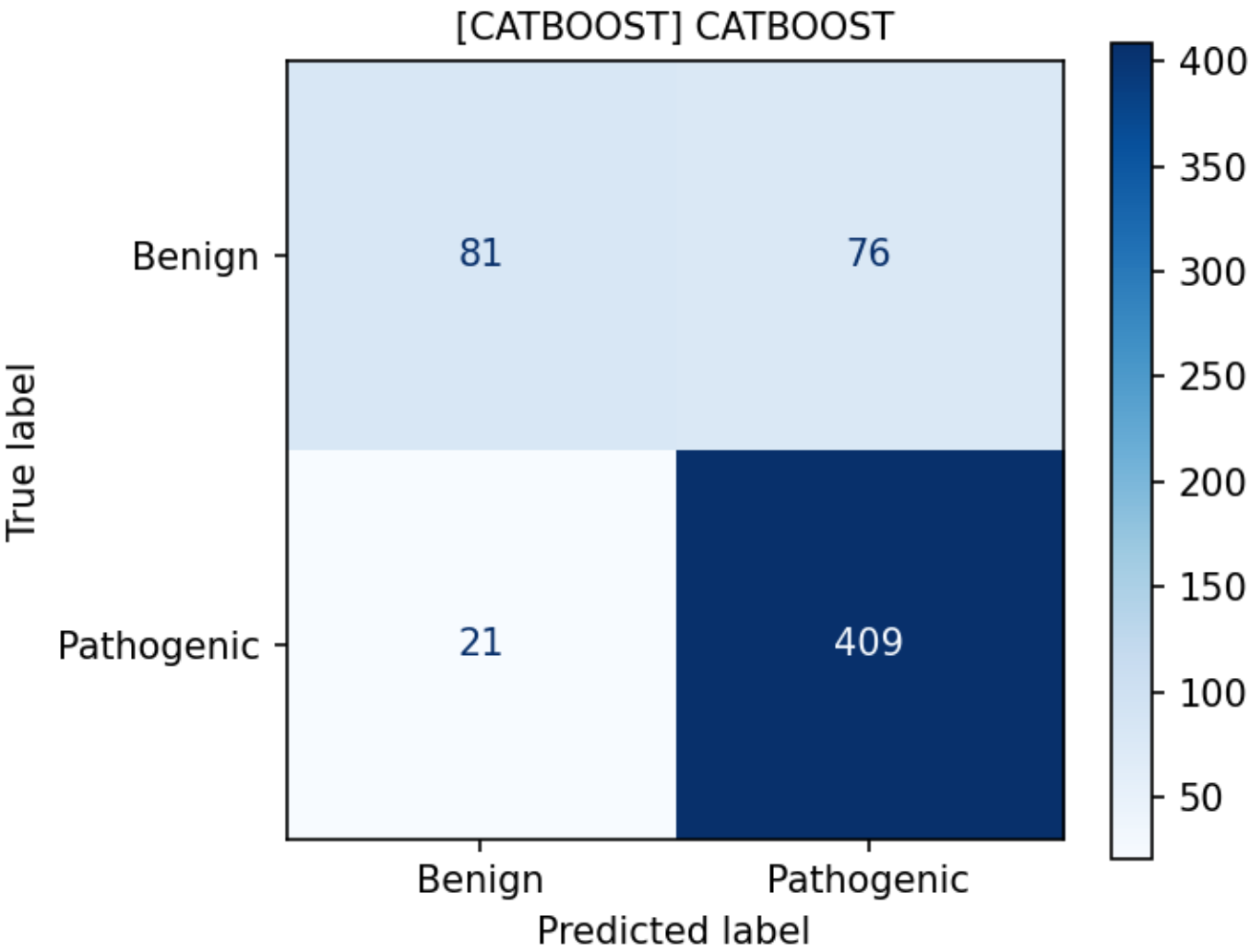
Master Xgb Le Cm



Master Svm Le Cm

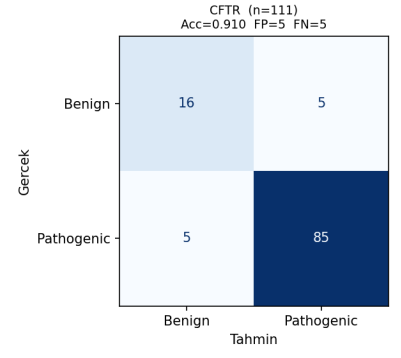
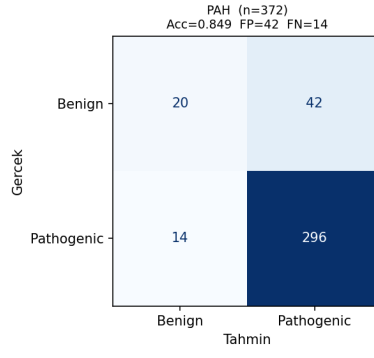
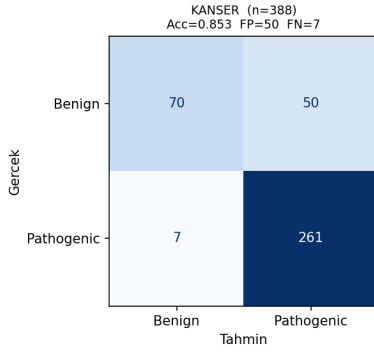


Master Catboost Cm



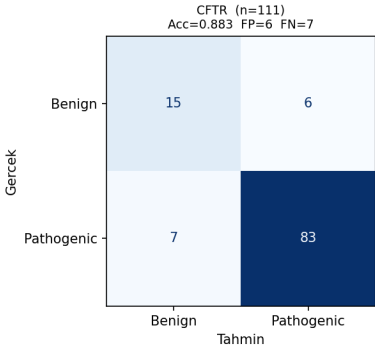
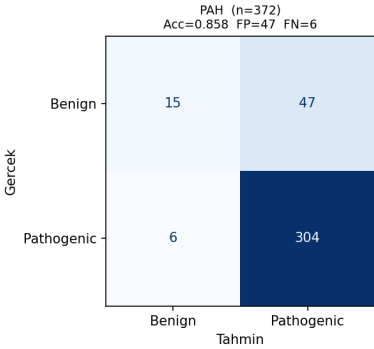
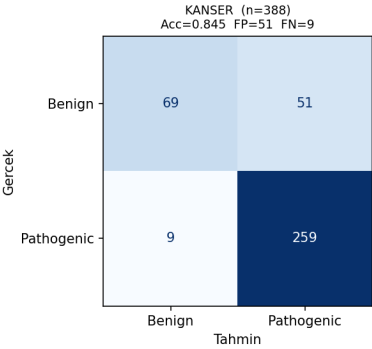
Panel Cm Lgbm Ohe

[OHE] LGBM_OHE -- Panel Bazli Confusion Matrix



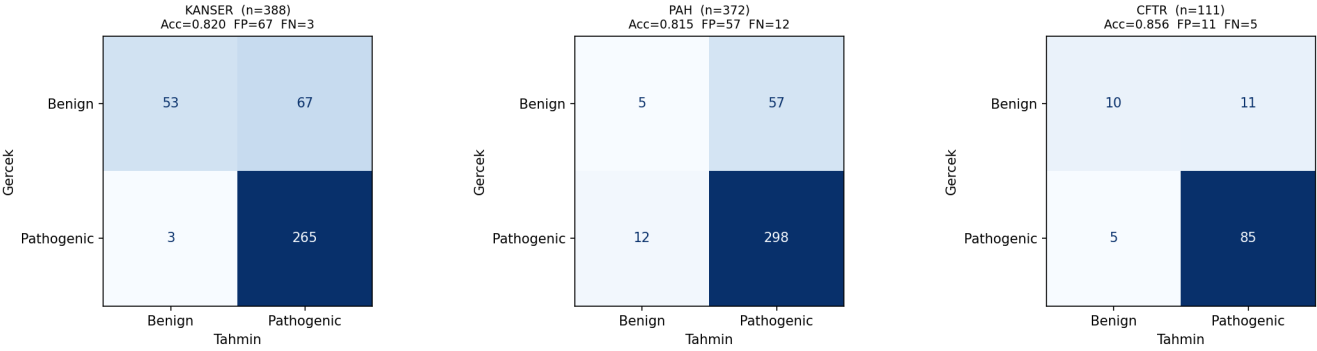
Panel Cm Xgb Ohe

[OHE] XGB_OHE -- Panel Bazli Confusion Matrix



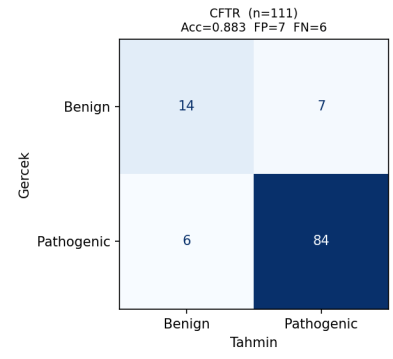
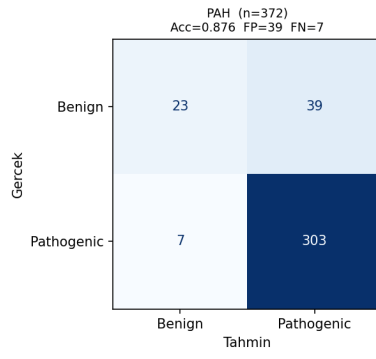
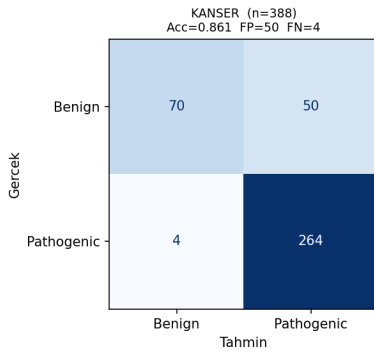
Panel Cm Svm Ohe

[OHE] SVM_OHE -- Panel Bazlı Confusion Matrix



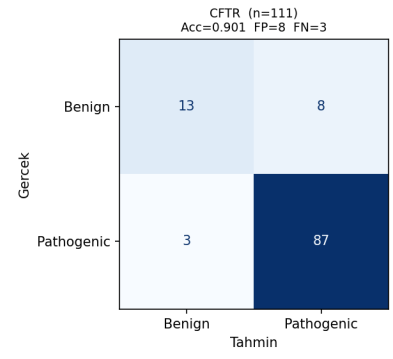
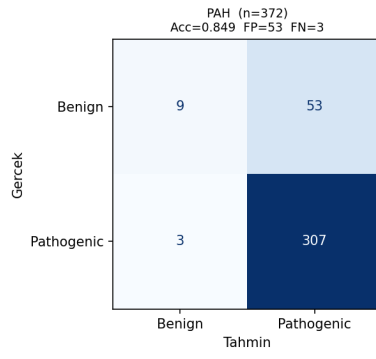
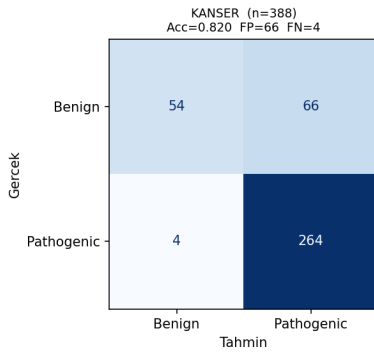
Panel Cm Lgbm Le

[LE] LGBM_LE -- Panel Bazli Confusion Matrix



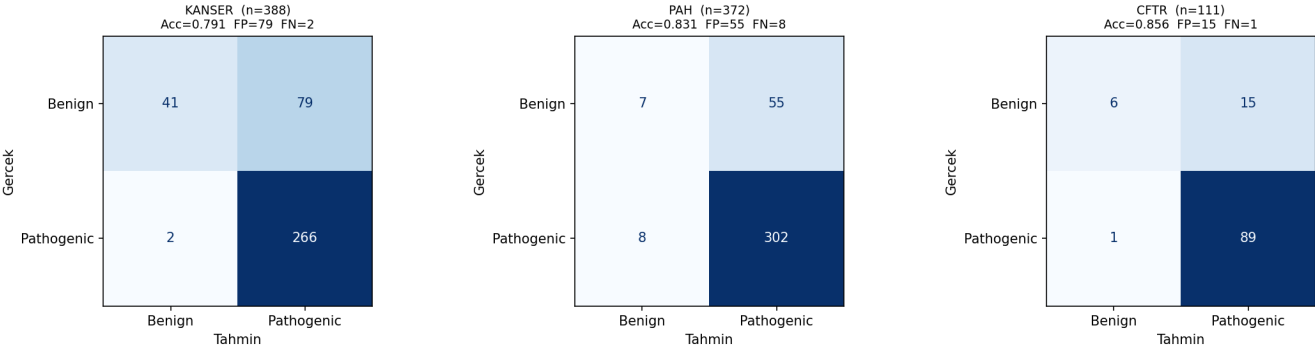
Panel Cm Xgb Le

[LE] XGB_LE -- Panel Bazli Confusion Matrix



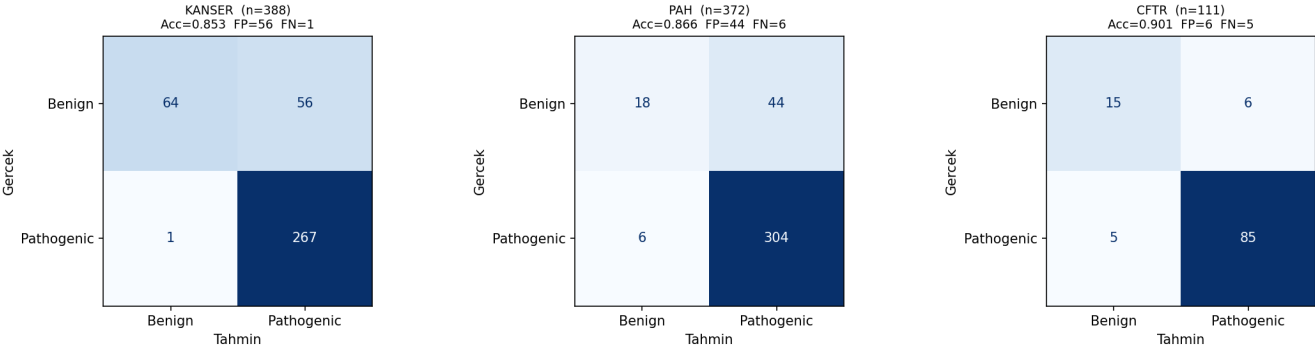
Panel Cm Svm Le

[LE] SVM_LE -- Panel Bazli Confusion Matrix



Panel Cm Catboost

[CATBOOST] CATBOOST -- Panel Bazli Confusion Matrix



Difficulty Distribution

